DEV SHETH

Brooklyn, NY

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Education

New York University Tandon School of Engineering **2025-2027**
Masters of Science, Cybersecurity 3.7/4
Relevant Coursework: Information Security & Privacy, Network Security, Penetration Testing & Vulnerability Analysis, Cloud Security, Application Security

M.S. Ramaiah University of Applied Sciences **2021-2025**
Bachelors of Technology, Computer Science and Engineering 8.5/10

Experience

Teaching Assistant – Cloud Security (CS-GY 9223), NYU Tandon School of Engineering ***(Sept 2026 - Dec 2026)***
- Support ~60 students in a graduate-level cloud security course under Prof. Catherine Dodge and Prof. Rima Tanash, covering AWS IAM, encryption, and cloud-native threat detection.
- Hold office hours and grade coursework, providing feedback on hands-on cloud security labs and assignments.

Business Development Intern- Healthcare Technology, Symphony Corporation, United States ***(Dec 24 - March 25)***
- Researched and identified 30+ potential healthcare organizations, building a robust pipeline for medical informatics outsourcing solutions.
- Maintained 30+ prospect interactions in ZOHO CRM, enabling smooth sales progression and reporting.
- Designed PostgreSQL data warehouse using 3NF and Kimball star schema, reducing query complexity for population health analytics and cutting report generation time by ~30%.

Projects

AWS IAM Security Monitor: Serverless Threat Detection Pipeline ***Feb 2026 - May 2026***
- Designed and implemented a cloud-native security monitoring pipeline on AWS using CloudTrail, EventBridge, Lambda, SNS, and CloudWatch.
- Configured multi-region CloudTrail logging, IAM roles, and S3 log storage to capture and centralize API activity across accounts.
- Built EventBridge rules to route high-risk IAM events to Lambda functions for real-time detection and automated remediation of privilege escalation.
- Delivered severity-based alerting via SNS and visualized incident activity through CloudWatch dashboards.

Secure Reference Monitor for Version Controlled File System ***October 2025***
- Implemented security kernel in RepyV2 enforcing mandatory access control with automatic file versioning.
- Mitigated TOCTOU vulnerabilities, race conditions, and concurrent access through thread synchronization.
- Developed 10+ attack test cases validating protections against unauthorized writes and version tampering.

Network Reconnaissance and Vulnerability Assessment ***November 2025***
- Conducted full-range network enumeration using Nmap across all 65,535 ports, identifying open services, OS fingerprints, and 5+ exploitable vulnerabilities on target hosts.
- Mapped attack surface by discovering 15+ subdomains via passive OSINT using Amass, exposing shadow IT and unmonitored external assets.

SCADA System Hardening & Kiosk-Mode Security Architecture ***December 2025***
- Architected a 4-layer security framework (RBAC, application whitelisting, RAM-based sandboxing, behavioral anomaly detection) for SCADA HMI workstations, preventing OS-level breakout.
- Validated framework resilience against 3 real-world ICS attack scenarios (Stuxnet, 2015 Ukrainian grid attack, Oldsmar breach), demonstrating mitigation at each attack vector.

Publications

Multimodal Sensor Fusion for Stress & Anxiety Detection ***April 2026***
- Developed an ML pipeline using wearable sensor data (accelerometer, EDA, skin temperature) to classify stress and anxiety in real time; decision tree model reached 93.77% accuracy. Published in Springer's Business Intelligence and Data Analytics (BIDA 2025) — doi.org/10.1007/978-3-032-07837-7_22

Certifications

- Machine Learning Specialization - DeepLearning.AI / Stanford University (Coursera, 2024).

Leadership & Affiliations

- Member, Google Developer Student Club, RUAS - Contributed to coding initiatives and peer learning (2022-2025).
- Lab Member, OSIRIS, NYU Center for Cybersecurity.

Technical Skills

- Programming & Scripting: Python, C, C++, Java, JavaScript, HTML, CSS.
- Security Tools: Kali Linux, Netcat, Amass, Sublist3r, ExifTool, SSH Tunneling, OSINT, Google Dorking.
- Penetration Testing: Metasploit, Nmap, Burp Suite, Hydra, Wireshark, Binary Exploitation, Shellcode Development, CVE Analysis.
- Development Tools & Databases: Git/GitHub, SQL/PostgreSQL/MongoDB, Pandas, NumPy, Matplotlib.
- Security Engineering: RBAC, Access Control, TOCTOU Prevention, Thread Synchronization, Network Segmentation.
- Cloud Security Tools: AWS, CloudTrail, Lambda, SNS, EventBridge, IAM, S3.